

Beginner's Introduction to Computing at CARC with JupyterHub

Version 0.1

Goals

- 1) Basic Linux Literacy
- 2) Using Compute Nodes with Slurm
- 3) Jupyterhub

I hope to spend about an hour on each with a 15 minute break.

Outline

- High Performance Computing Overview
- Logging in
- The BASH Shell
- The Slurm Job Scheduler
- Transferring data to and from CARC
- Accessing software and the module system
- Conda environments
- Jupyterhub

High Performance Computing

- What is high performance computing?
 - Really just means something that is a lot more powerful than your desktop or laptop.
 - Hardware:
 - That might mean more and faster processors to do the calculations more quickly (eg 400 CPUs instead of 4)
 - More RAM so you can work on bigger problems (3,000 GB instead of 8)
 - Bigger file systems so you can process larger datasets
 - More and bigger GPUs to accelerate your computations (12 GPUs at a time instead of 1)
 - People:
 - Someone else to manage the systems and keep them running and secure
 - Someone to answer your questions and help with problems

Logging into Wheeler



First login to the Linux **workstation** in front of you.

Use your CARC username and password.

Keven, Tannor, Viacheslav (Slava), and Jose can help you login if you have trouble.

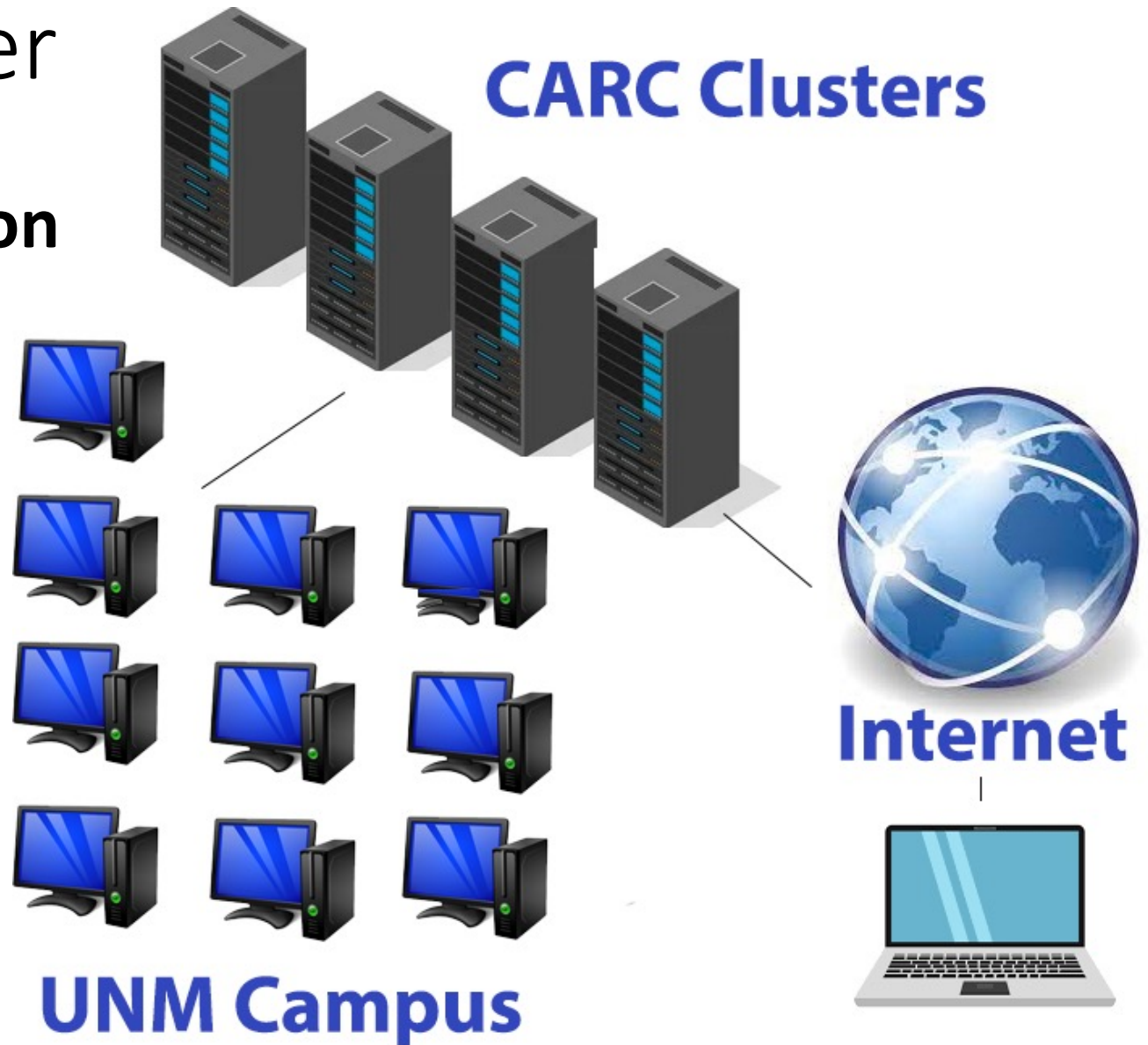
This is an “important step” so don’t let me move on until you have logged in

Logging into Wheeler

First login to the **workstation** in front of you.

You will always login to CARC cluster remotely.

These clusters don't even have monitors.



Logging into Wheeler

We are going to use a program called [secure shell](#).

[Secure shell \(ssh\)](#) is now built into every major operating system (Windows, OSX, and Linux).

You don't need third party programs like putty anymore.

Logging into Wheeler



```
ssh vanilla@wheeler.alliance.unm.edu
```

Should prompt you for a password...

Don't let me move on until you are able to login.

Logging into Wheeler

```
Matthew — ssh wheeler — 82x27
last login: Tue Jun 14 14:47:24 2022 from fricke.co.uk
-----
Welcome to Wheeler

Be sure to review the "Acceptable Use" guidelines posted on the CARC website.

For assistance using this system email help@carc.unm.edu.

Tutorial videos can be accessed through the CARC website: Go to
http://carc.unm.edu, select the "New Users" menu and then click
"Introduction to Computing at CARC".

Warning: By default home directories are world readable. Use the chmod command
to restrict access.

Don't forget to acknowledge CARC in publications, dissertations, theses and
presentations that use CARC computational resources:

"We would like to thank the UNM Center for Advanced Research Computing,
supported in part by the National Science Foundation, for providing the
research computing resources used in this work."

Please send citations to publications@carc.unm.edu.
-----
Starting SSH Key Agent...
agent pid 19486
fricke@wheeler:~ $
```

Please enter the following command



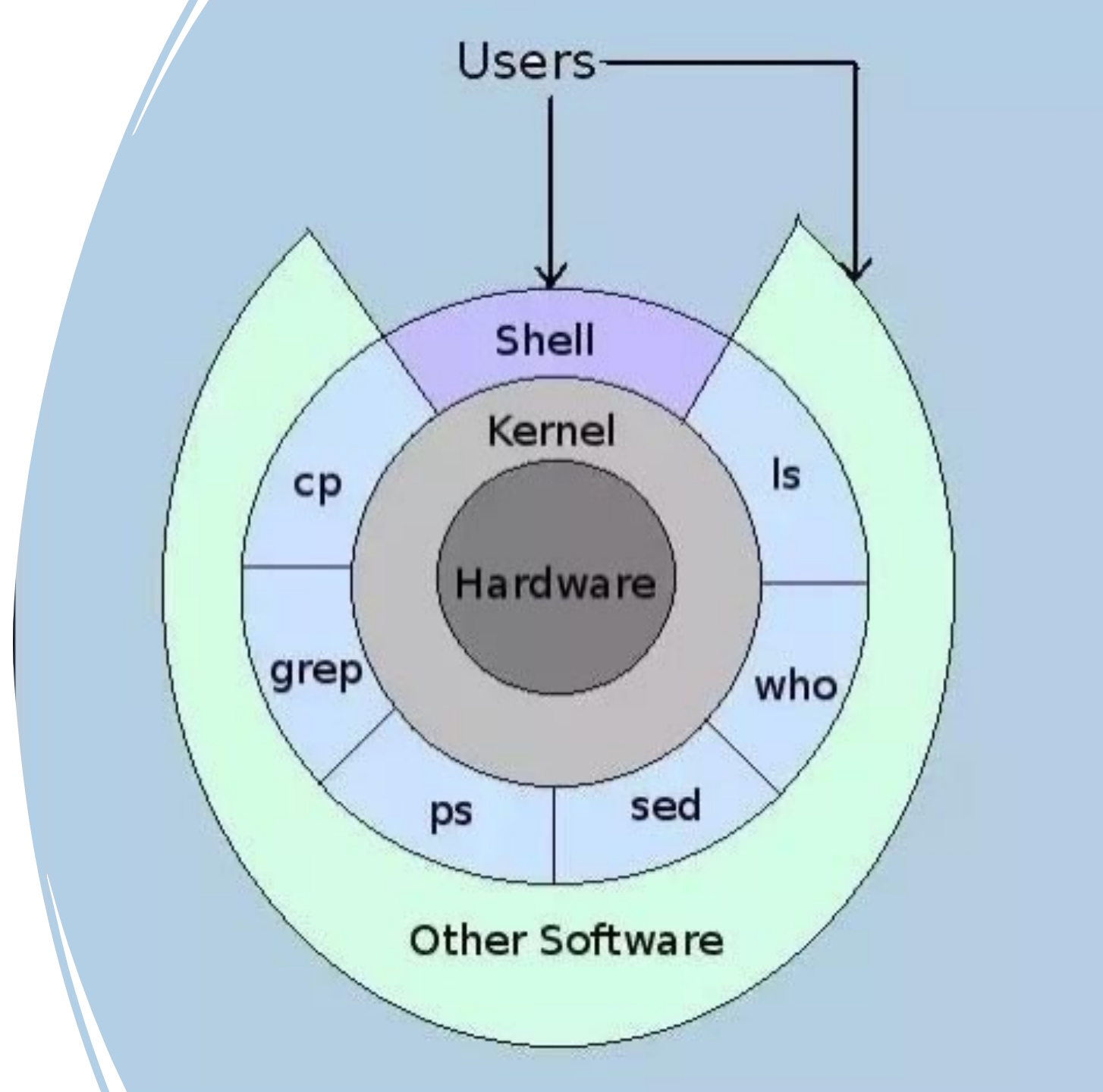
```
cp -r /projects/shared/workshops/beginner/mystuff ~/
```

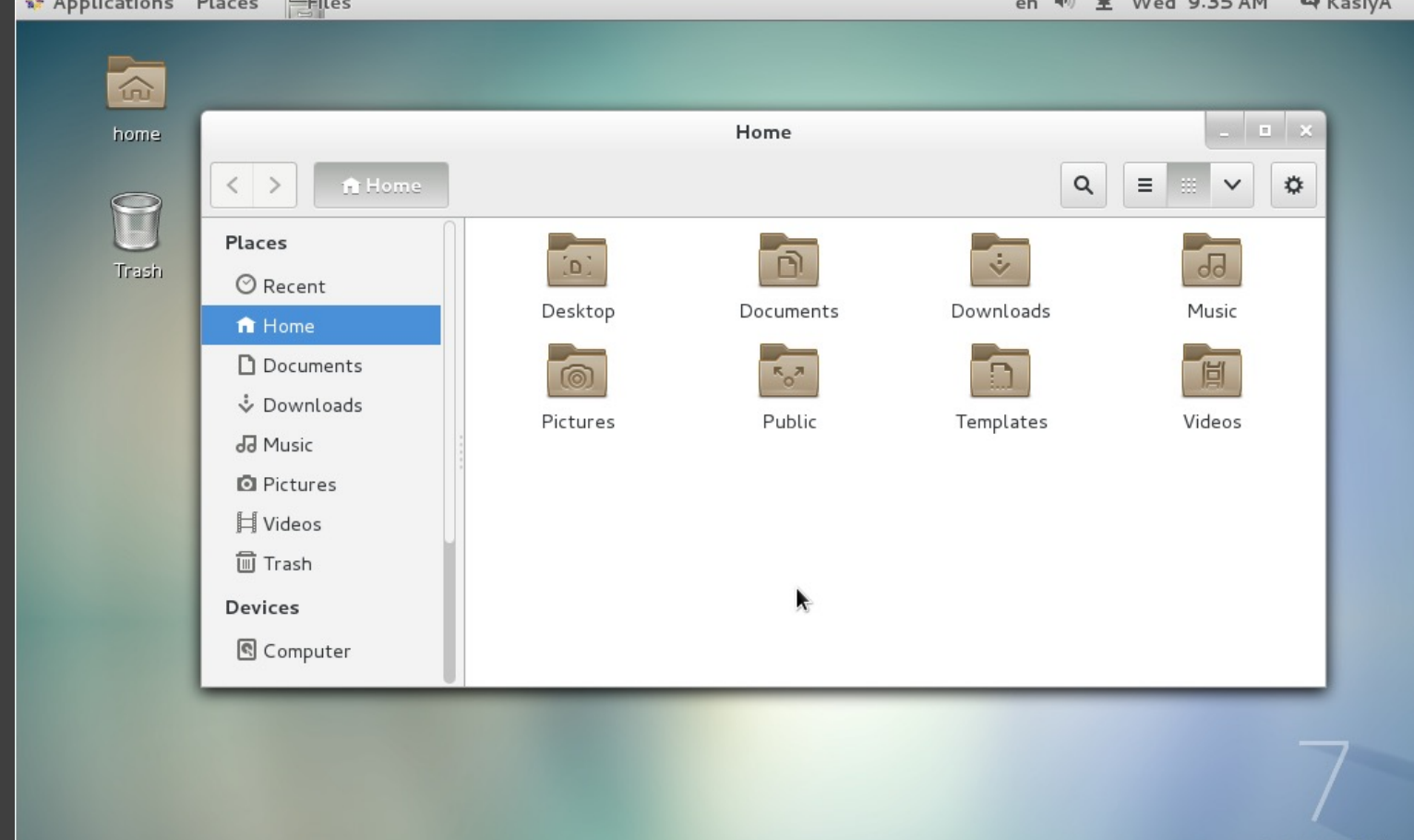
We will come help you if you have any trouble.

(Later I will go over what this command does)

Linux and the BASH Shell

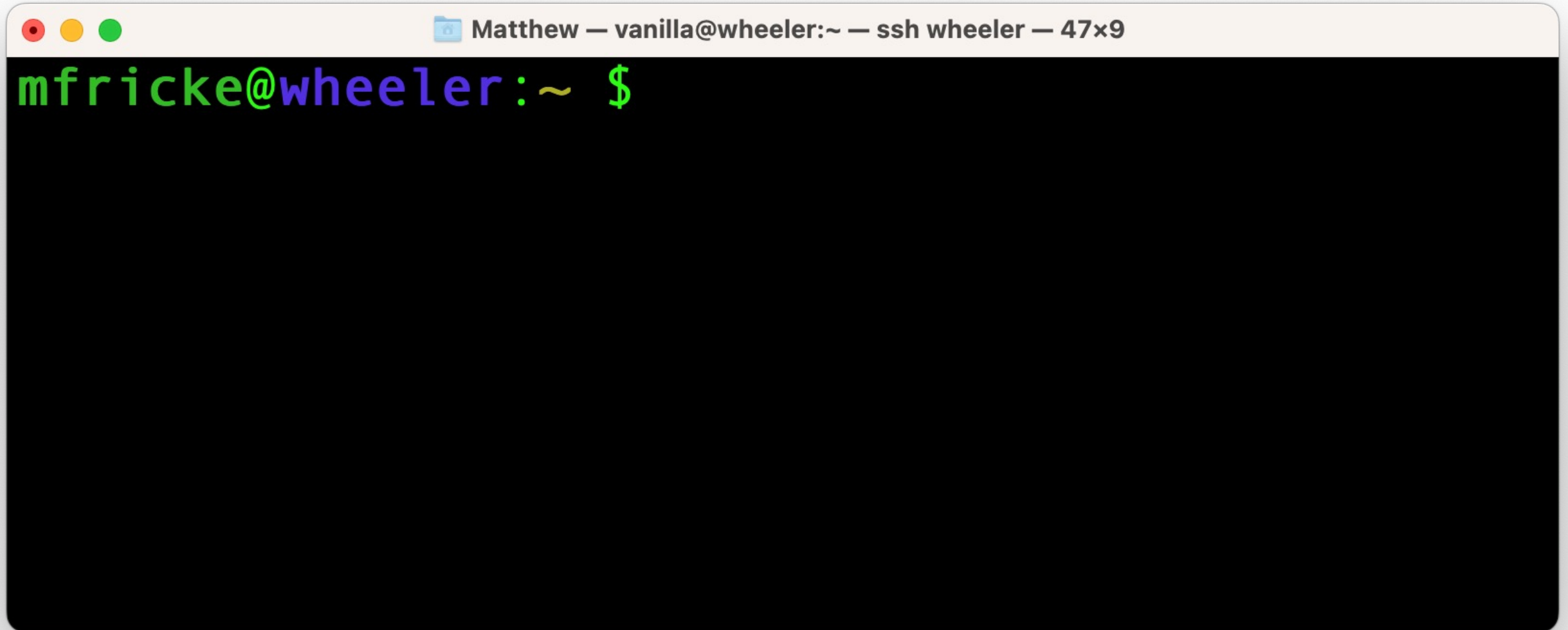
- The Kernel manages access to the hardware in a computer.
- An Operating System (OS) is the Kernel plus useful programs provided by the OS.
- The “shell” is the outermost layer of the OS.
- It is where the user interacts with the OS.





Graphical Shells (GUIs)

Logging into Wheeler



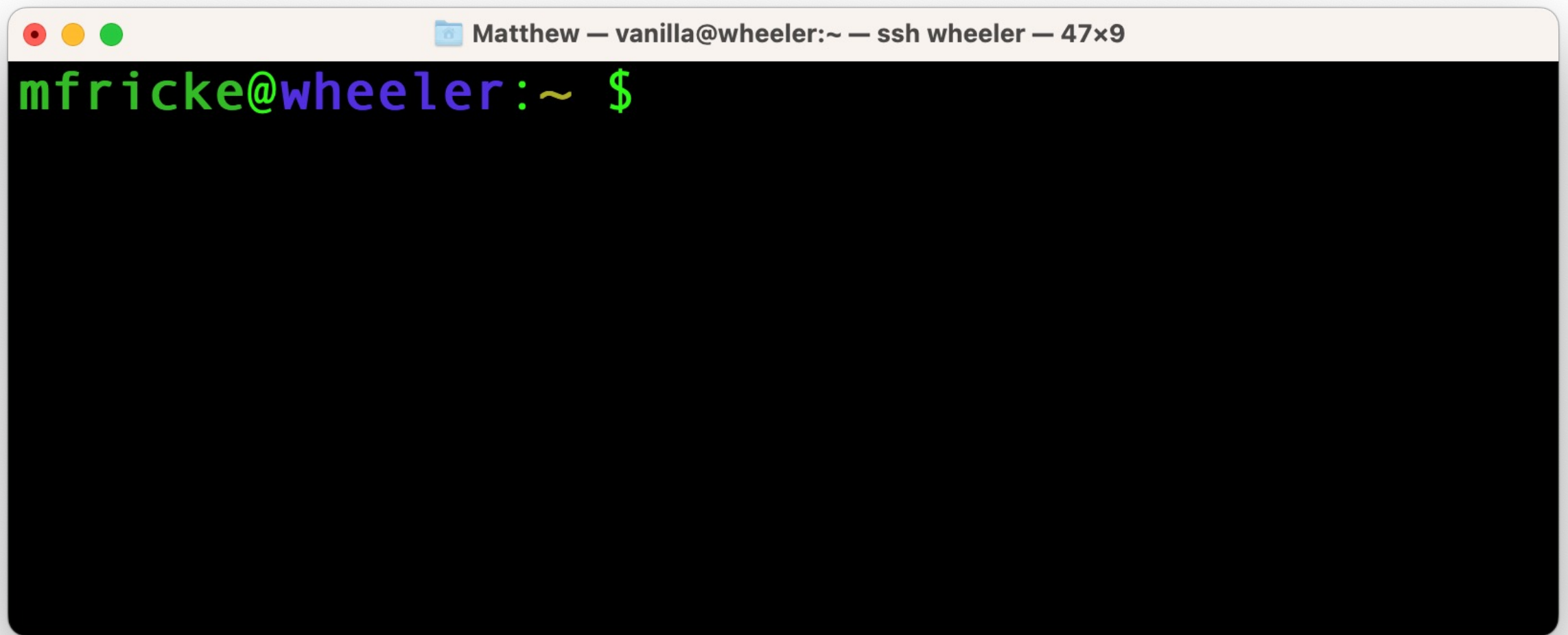
Linux and the BASH Shell

The word "LINUX" is rendered in a stylized, blocky font using only vertical and horizontal lines, giving it a digital or circuit-like appearance. Each letter is composed of multiple parallel lines, creating a sense of depth and texture. The letters are white and stand out against the black background.

WHERE THERE IS A SHELL, THERE IS A WAY!

The Borne-Again Shell (BASH)

Written in 1976 by Stephen Bourne for UNIX version 7.





A terminal window with a title bar that reads "Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9". The terminal content shows the prompt "mfricke@wheeler:~ \$" in green and blue. A blue arrow points from the word "Username" below to the "mfricke" part of the prompt.

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9  
mfricke@wheeler:~ $  
↑  
Username
```

Understanding the BASH prompt...



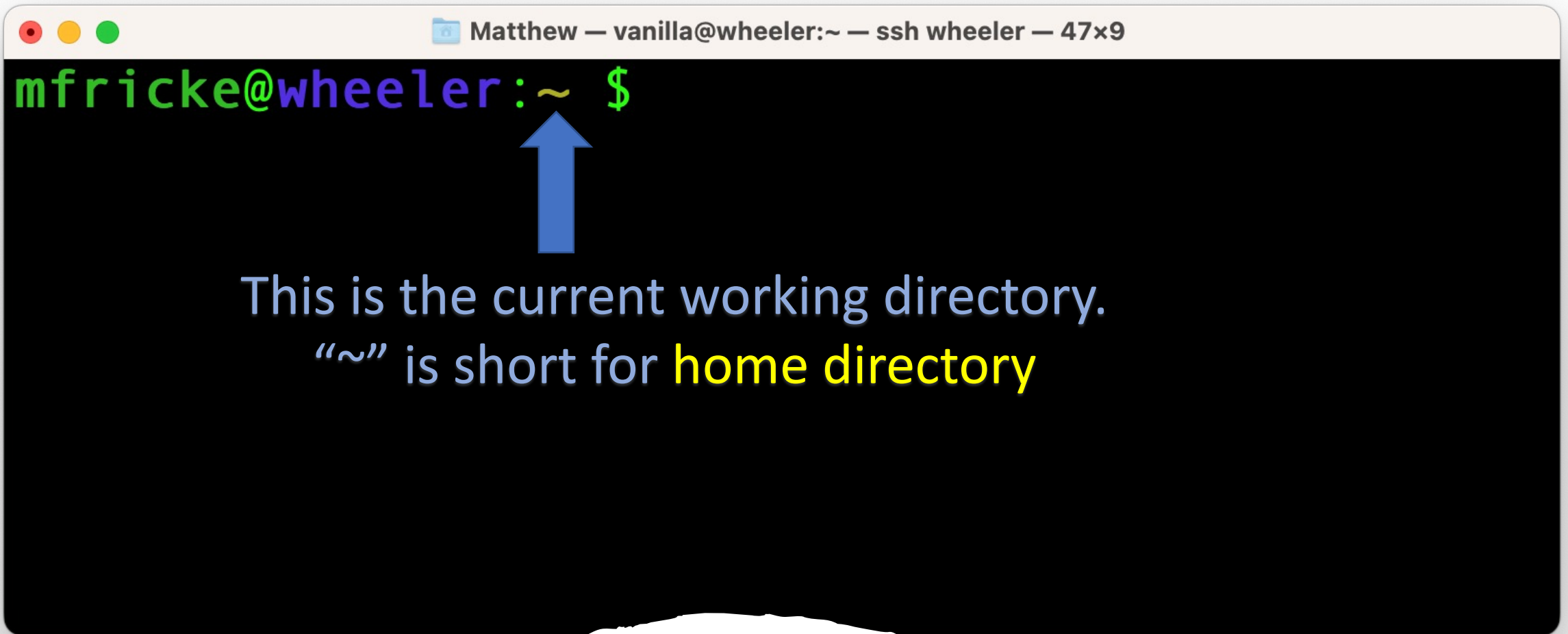
Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9

```
mfricke@wheeler:~ $
```

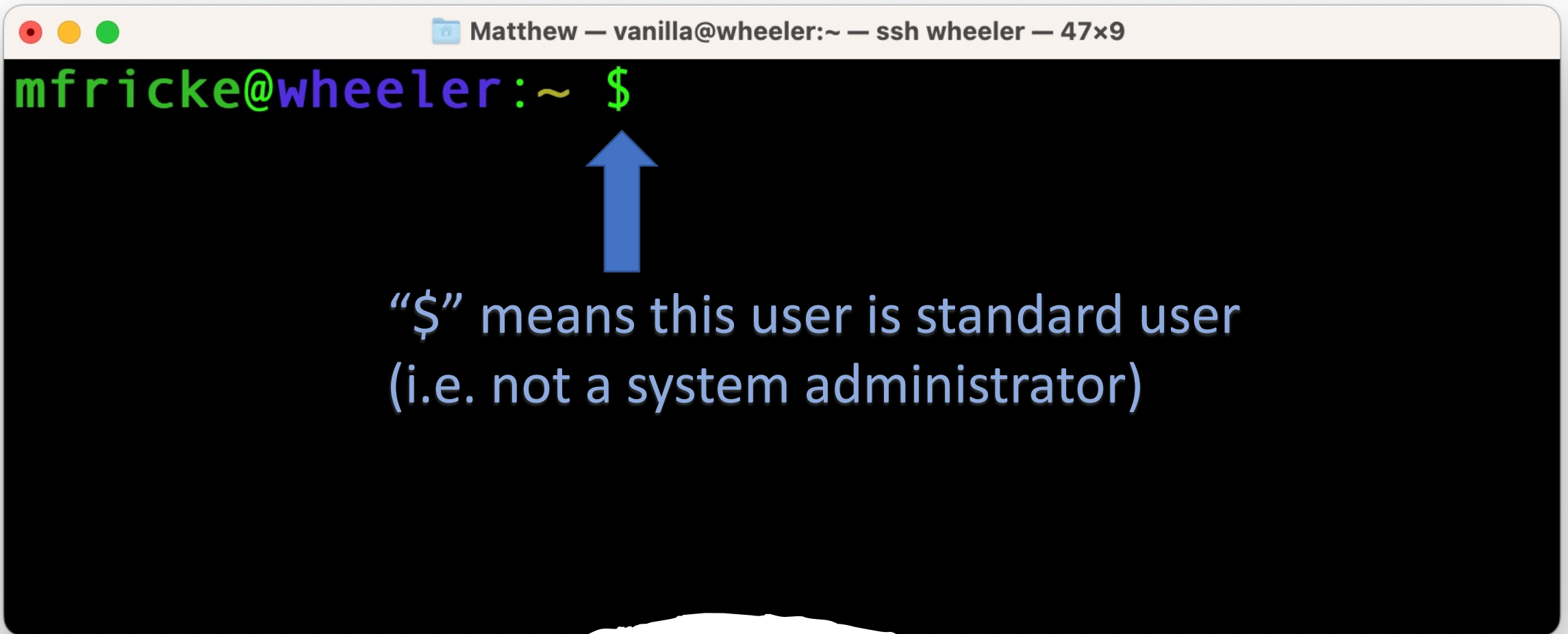
↑ ↑

Username Hostname

Understanding the BASH prompt...



Understanding the BASH prompt...

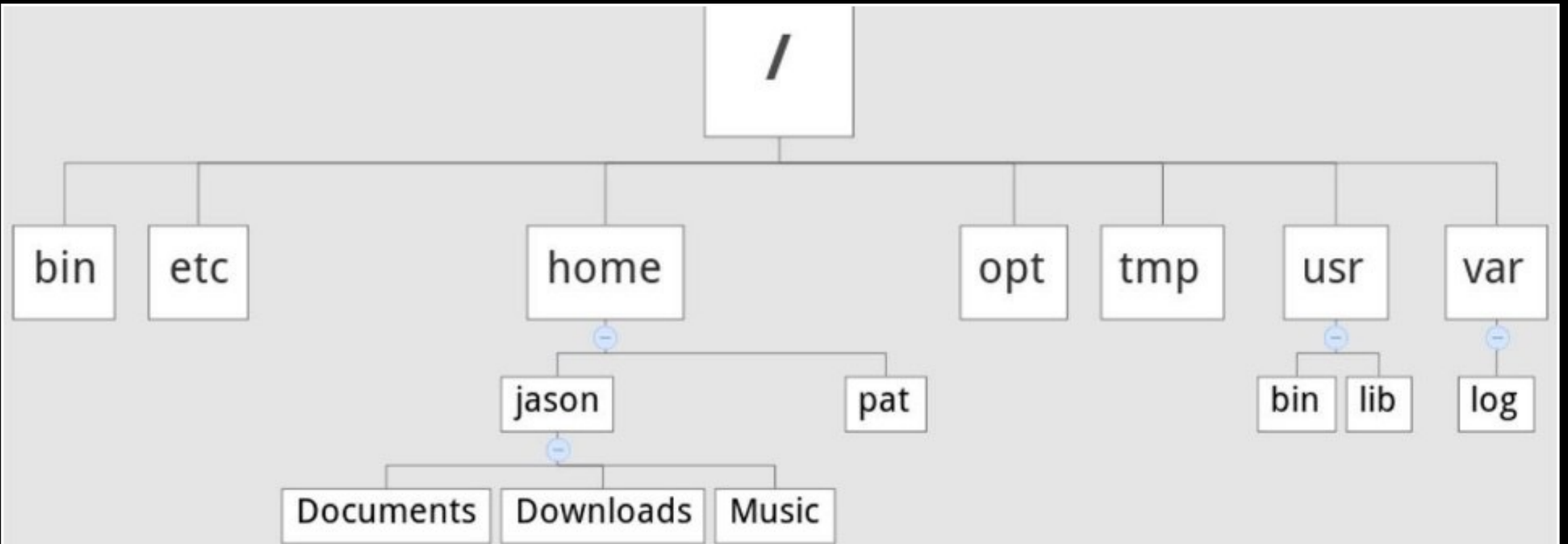


A terminal window with a title bar that reads "Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9". The terminal content shows a BASH prompt "mfricke@wheeler:~ \$" where "mfricke@" is green, "wheeler:" is blue, "~" is yellow, and "\$" is green. A blue arrow points from the text below to the "\$" character. The text below the arrow reads: "\$" means this user is standard user (i.e. not a system administrator).

```
mfricke@wheeler:~ $
```

“\$” means this user is standard user
(i.e. not a system administrator)

Understanding the BASH prompt...



Example Filesystem Tree

```
[vanilla@wheeler ~]$ pwd  
/users/vanilla  
[vanilla@wheeler ~]$
```

Figuring out where you are in the
filesystem...

```
[vanilla@wheeler ~]$ ls  
mystuff  wheeler-scratch  
[vanilla@wheeler ~]$
```

Figuring out where you are in the
filesystem...

```
[vanilla@wheeler ~]$ tree
```

```
.
├── mystuff
│   ├── myfile1
│   └── myfile2
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

```
2 directories, 2 files
```

```
[vanilla@wheeler ~]$
```

Figuring out where you are in
the filesystem...

```
[vanilla@wheeler ~]$ tree
```



```
.  
├── mystuff  
│   ├── myfile1  
│   └── myfile2  
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

This . means the current directory

```
2 directories, 2 files
```

```
[vanilla@wheeler ~]$
```

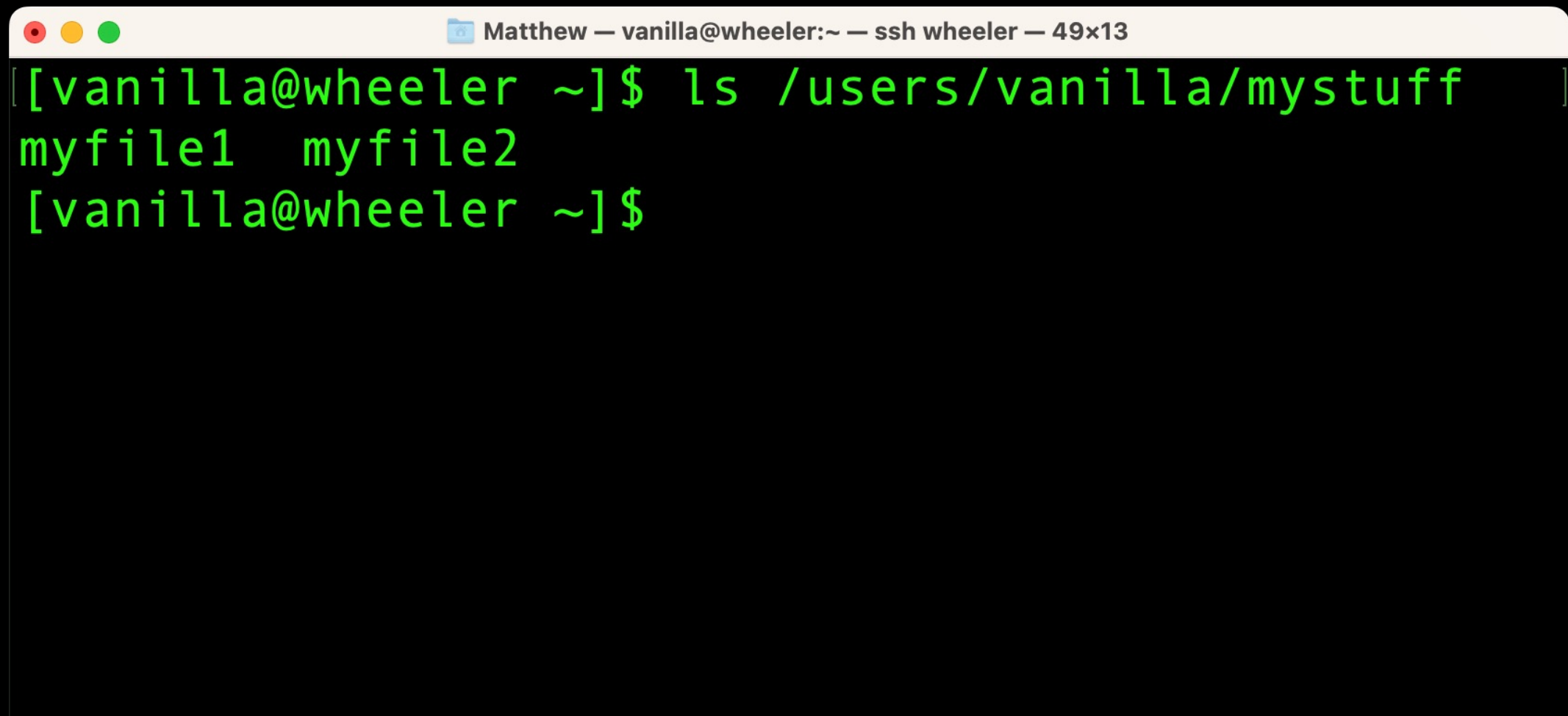
Figuring out where you are in
the filesystem...

“Absolute” paths vs “relative” paths

- A path is a list of directories and/or files. It is a path through the directory tree that tells one how to get somewhere in the filesystem.
- An absolute path tells one how to get to the destination from starting from the root of the filesystem. E.g “/users/vanilla/mystuff/”
- A relative path specifies how to get there *starting from the current working directory*. E.g vanilla/mystuff/

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 49x13  
[vanilla@wheeler ~]$ ls mystuff/  
myfile1  myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...

A terminal window with a title bar containing three colored circles (red, yellow, green) and the text "Matthew — vanilla@wheeler:~ — ssh wheeler — 49x13". The terminal content shows a command prompt "[vanilla@wheeler ~]\$", followed by the command "ls /users/vanilla/mystuff", and the output "myfile1 myfile2". The prompt "[vanilla@wheeler ~]\$" appears again on the next line.

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 49x13  
[vanilla@wheeler ~]$ ls /users/vanilla/mystuff  
myfile1  myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls ./mystuff/
myfile1  myfile2
[vanilla@wheeler ~]$ ls ~/mystuff/
myfile1  myfile2
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -a
.          .modulesbeginenv
..         mystuff
.addressbook .oracle_jre_usage
.addressbook.lu .pinerc
.bashrc      .pki
.cache       .rhosts
.comsol      .shosts
.config      .spack
.flexlmrc    .ssh
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -l
total 4
drwxr-xr-x 2 vanilla users 4096 Jun 14 22:05 mystuff
lrwxrwxrwx 1 vanilla users 24 Jun 14 21:20 wheeler-scratch ->
/wheeler/scratch/vanilla
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -l mystuff/  
total 473704  
-rw-r--r-- 1 vanilla users 483165473 Jun 14 23:20 myfile1  
-rw-r--r-- 1 vanilla users      0 Jun 14 22:05 myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -lh mystuff/
total 463M
-rw-r--r-- 1 vanilla users 461M Jun 14 23:20 myfile1
-rw-r--r-- 1 vanilla users    0 Jun 14 22:05 myfile2
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ du -s
```

```
499704 .
```

```
[vanilla@wheeler ~]$ du -sh
```

```
488M .
```

```
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	24G	0	24G	0%	/dev
tmpfs	24G	64K	24G	1%	/dev/shm
tmpfs	24G	968M	23G	5%	/run
tmpfs	24G	0	24G	0%	/sys/fs/cgroup
/dev/mapper/centos-root	930G	567G	363G	61%	/
/dev/sdc2	836G	72G	764G	9%	/tmp
/dev/md126p1	2.0G	333M	1.7G	17%	/boot
172.17.2.254:/mnt/wheeler-scratch	37T	28T	8.7T	77%	/wheeler/scratch
172.17.2.255:/mnt/wheeler-scratch2	37T	28T	9.0T	76%	/wheeler/scratch2
beegfs_nodev	110T	51T	60T	46%	/carc/scratch
chama:/home/homes	65T	36T	30T	55%	/users
chama:/home/carc_projects	65T	36T	30T	55%	/projects

Figuring out where you going...

```
[vanilla@wheeler ~]$ quota -s
Disk quotas for user vanilla (uid 659):
    Filesystem    space   quota   limit   grace   files   quota   limit   grace
chama:/home/homes
                488M    100G    200G           315    4295m    4295m
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ stat mystuff/myfile1
```

```
  File: 'mystuff/myfile1'  
  Size: 483165473 Blocks: 947408      IO Block: 65536   regular file  
Device: 28h/40d Inode: 9232782834205560540  Links: 1  
Access: (0644/-rw-r--r--)  Uid: (  659/  vanilla)   Gid: (  100/   users)  
Access: 2022-06-14 22:05:27.503289000 -0600  
Modify: 2022-06-14 23:20:26.945918000 -0600  
Change: 2022-06-14 23:20:48.754917000 -0600  
 Birth: -
```

Figuring out what you've got...

```
[vanilla@wheeler ~]$ find -name myfile2  
./mystuff/myfile2
```

```
[vanilla@wheeler ~]$ find -name "myfile*"  
./mystuff/myfile1  
./mystuff/myfile2  
./mystuff/myfile3  
./mystuff/myfile0
```



Wildcard

Figuring out what you've got...

```
[vanilla@wheeler ~]$ cd mystuff/  
[vanilla@wheeler ~/mystuff]$
```

Use the tab key to autocomplete

Going somewhere new...

Now it is your turn...



- For this path:
`/projects/shared/workshops/beginner/vecadd`

- What are the names of the files in that directory?
- When were they last modified?
- How large are the files?

You can find this information with the `ls` command.

Now it is your turn...



- For this path:
`/projects/shared/workshops/beginner/vecadd`

Now “cd” into that directory using <tab> autocomplete.

- Now you know how to find your way around filesystems using bash
- Let's see how to modify the filesystem.
- In bash to move a file we use the `mv` command.
- To copy a file it is `cp`.
- To copy files from CARC to a personal computer use `scp` or `rsync`.

```
[vanilla@wheeler beginner]$ pwd
/projects/shared/workshops/beginner
[vanilla@wheeler beginner]$ cd ~
[vanilla@wheeler ~]$ pwd
/users/vanilla
[vanilla@wheeler ~]$
```

First return to your home
directory...

```
[vanilla@wheeler ~]$ cd mystuff
[vanilla@wheeler ~/mystuff]$ mv myfile1 myfile0
[vanilla@wheeler ~/mystuff]$ ls
myfile0  myfile2  myfile3
[vanilla@wheeler ~/mystuff]
```

Modifying the filesystem...
moving a file.

```
[vanilla@wheeler ~/mystuff]$ cp myfile0 myfile1  
[vanilla@wheeler ~/mystuff]$
```



Source



Destination

```
[vanilla@wheeler ~/mystuff]$ ls  
myfile0  myfile1  myfile2  myfile3  
[vanilla@wheeler ~/mystuff]$
```

Modifying the filesystem...
copying a file.

```
[vanilla@wheeler ~]$ cp -r mystuff mystuff2  
[vanilla@wheeler ~]$
```



Source



Destination

```
[vanilla@wheeler ~]$ ls  
mystuff    mystuff2    wheeler-scratch
```

Copying a whole directory tree...

```
Lycaon:~ matthew$ scp vanilla@wheeler.alliance.unm.edu:~/mystuff/myfile3 Desktop/
```



Source



Destination

```
(vanilla@wheeler.alliance.unm.edu) Password:  
myfile3          100%  40      2.0KB/s   00:00
```

Copying data to a personal
computer from CARC...

```
Lycaon:~ matthew$ scp -r vanilla@wheeler.alliance.unm.edu:~/mystuff Desktop/
```



Source



Destination

```
(vanilla@wheeler.alliance.unm.edu) Password:
```

myfile1	100%	1024KB	6.5MB/s	00:00
myfile2	100%	2048KB	382.5KB/s	00:05
myfile3	100%	40	3.2KB/s	00:00
myfile0	100%	1024KB	8.8MB/s	00:00

Copying data to a personal
computer from CARC...

```
Lycaon:~ matthew$ scp -r Desktop/mystuff vanilla@wheeler.alliance.unm.edu:~/
```



Source



Destination

```
(vanilla@wheeler.alliance.unm.edu) Password:
```

myfile1	100%	1024KB	591.5KB/s	00:01
myfile0	100%	1024KB	2.0MB/s	00:00
myfile2	100%	2048KB	2.1MB/s	00:00
myfile3	100%	40	2.1KB/s	00:00

To copy from a personal
computer to CARC...

```
ssh vanilla@wheeler.alliance.unm.edu
```

Log back into wheeler...

```
[vanilla@wheeler ~]$ file mystuff/myfile0  
mystuff/myfile0: data
```

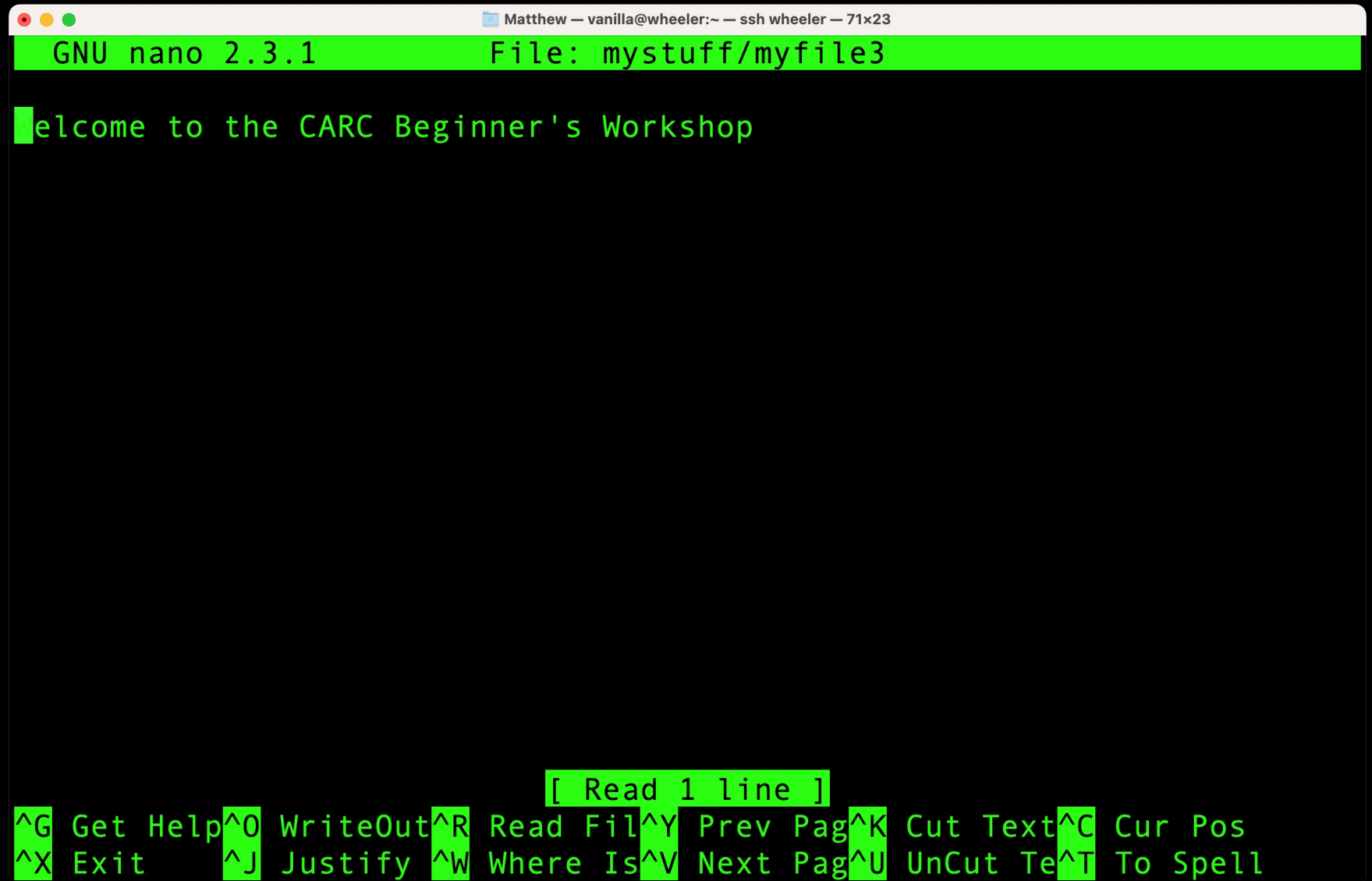
```
[vanilla@wheeler ~]$ file mystuff/myfile3  
mystuff/myfile3: ASCII text
```

Figuring out file types ...

```
[vanilla@wheeler ~]$ cat mystuff/myfile3  
Welcome to the CARC Beginner's Workshop
```

Text files ...

```
[vanilla@wheeler ~]$ nano mystuff/myfile3
```



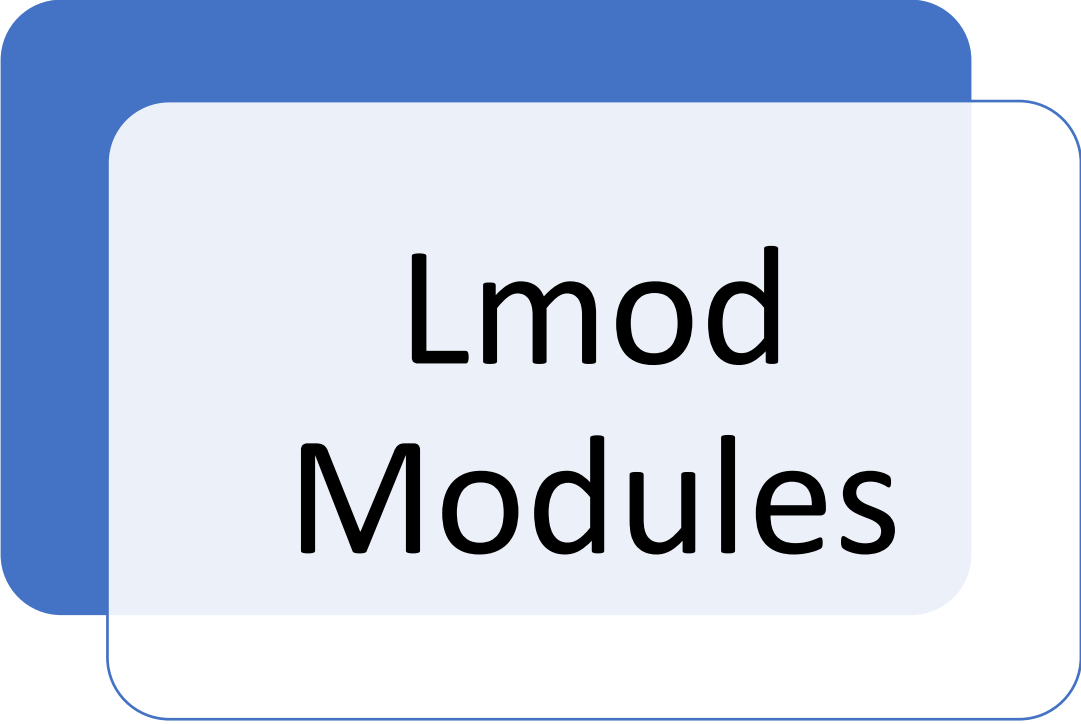
```
Matthew — vanilla@wheeler:~ — ssh wheeler — 71x23
GNU nano 2.3.1      File: mystuff/myfile3

Welcome to the CARC Beginner's Workshop

[ Read 1 line ]
^G Get Help ^O WriteOut ^R Read Fil ^Y Prev Pag ^K Cut Text ^C Cur Pos
^X Exit     ^J Justify  ^W Where Is ^V Next Pag ^U UnCut Te ^T To Spell
```

15 Minute Break

Software Access



Lmod
Modules

The diagram consists of a dark blue rounded rectangle with a light blue rounded rectangle nested inside it. The text "Lmod Modules" is centered within the light blue rectangle.



Conda

The diagram consists of a dark blue rounded rectangle with a light blue rounded rectangle nested inside it. The text "Conda" is centered within the light blue rectangle.

```
[vanilla@wheeler ~]$ module spider matlab
```

```
-----
```

```
-
```

```
  matlab:
```

```
-----
```

```
-
```

```
  Versions:
```

```
    matlab/R2017a
```

```
    matlab/R2018b
```

```
    matlab/R2019a
```

```
    matlab/R2020a
```

```
    matlab/R2021a
```

Getting access to software...

```
[vanilla@wheeler ~]$ module load matlab/R2021a
Lmod has detected the following error: Matlab may only be run on compute
nodes. wheeler is not a compute node. Exiting...
While processing the following module(s):
  Module fullname  Module Filename
  -----
matlab/R2021a      /opt/local/modules/matlab/R2021a.lua
```

Getting access to software...

```
[vanilla@wheeler ~]$ module load matlab/R2021a
Lmod has detected the following error: Matlab may only be run on compute
nodes. wheeler is not a compute node. Exiting...
While processing the following module(s):
  Module fullname  Module Filename
  -----
matlab/R2021a      /opt/local/modules/matlab/R2021a.lua
```

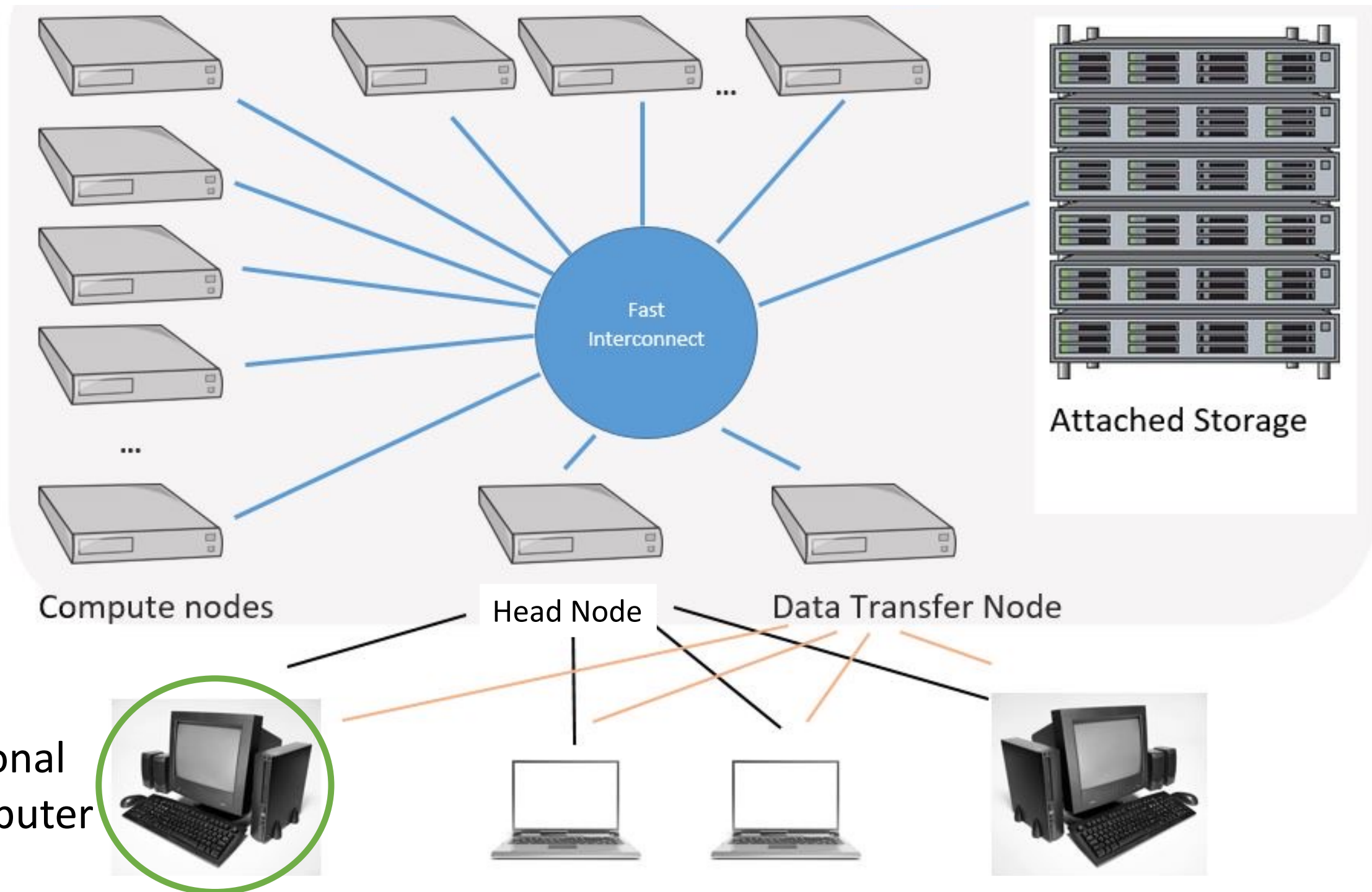
What is a compute node?

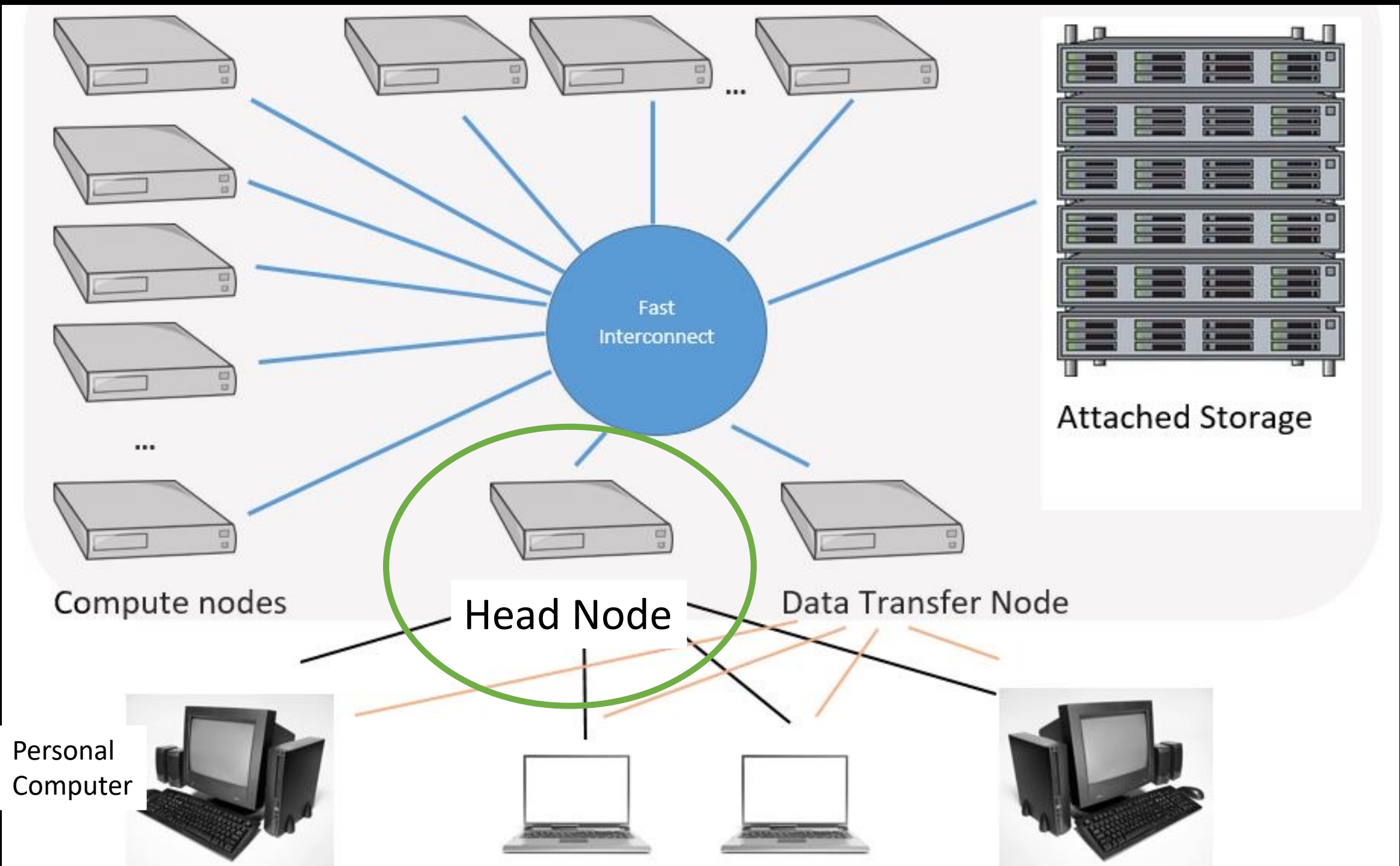
Getting access to software...

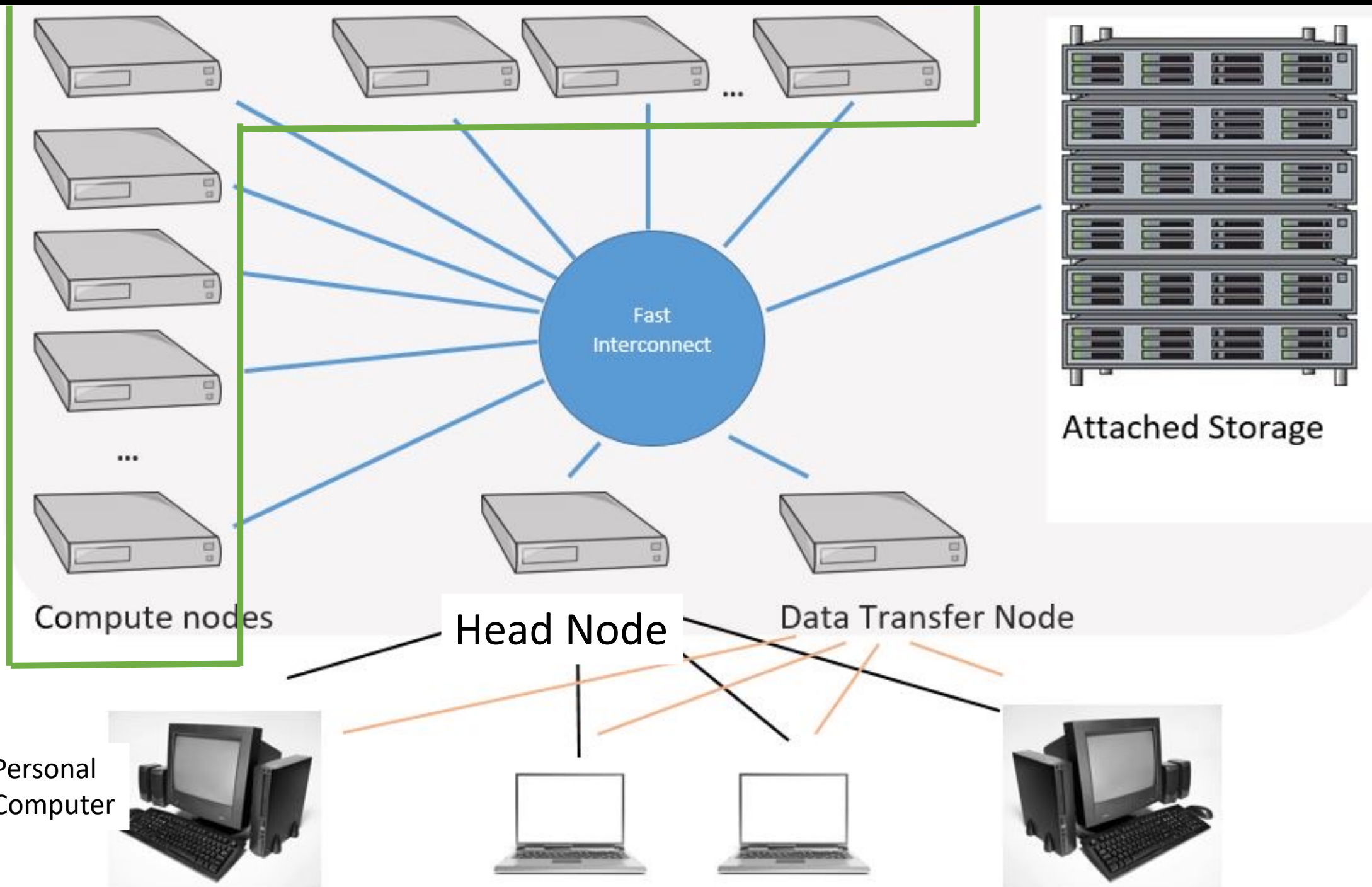
HPC Cluster



Personal
Computer









Never run computations on the head node

Always use compute nodes

```
[vanilla@wheeler ~]$ qgrok
```

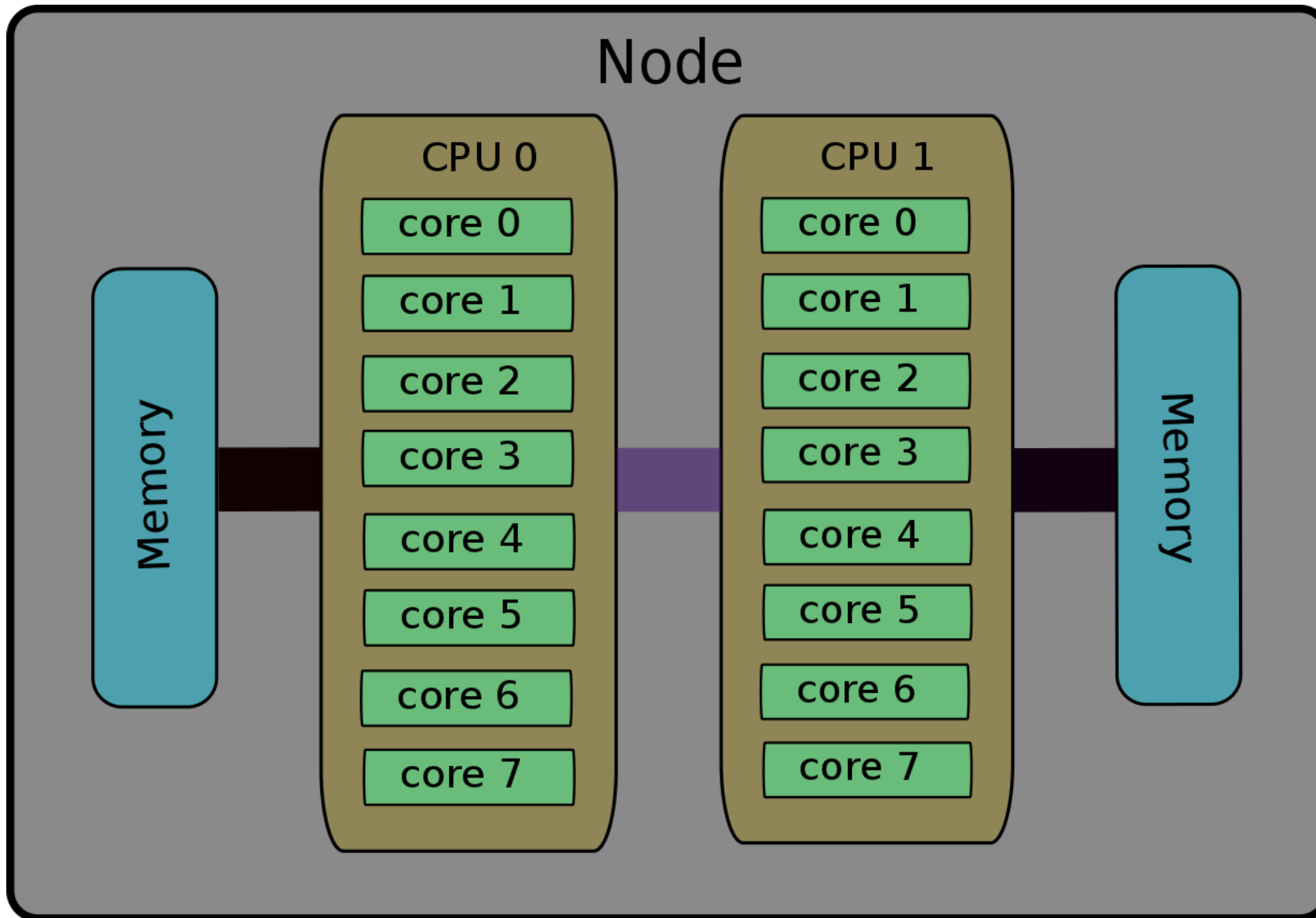
queues	free	busy	offline	jobs	nodes	CPUs
-----	-----	-----	-----	-----	-----	-----
normal	0	299	1	97	300	2400
debug	4	0	0	0	4	32
totals:	4	299	1	97	304	2432

Compute nodes and partitions...

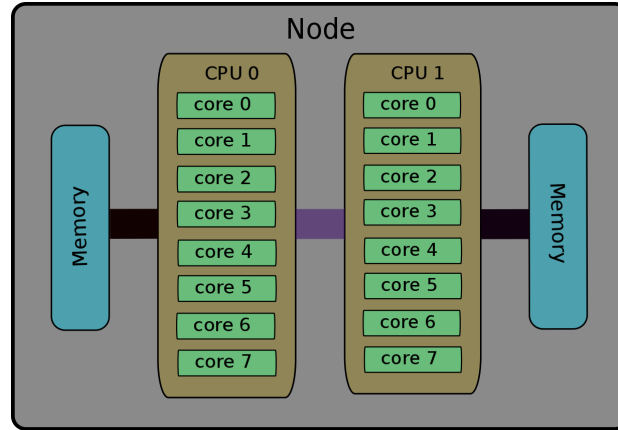
```
qmfricke@hopper:~ $ qgrok
```

queues	free	busy	offline	jobs	nodes	CPUs
-----	-----	-----	-----	-----	-----	-----
general	12	0	0	0	12	384
condo	29	1	7	0	37	1184
bugs	2	0	0	0	2	64
pcnc	2	0	0	0	2	64
pathogen	0	1	0	1	1	32
tc	7	0	3	0	10	320
gold	2	0	0	0	2	64
fishgen	1	0	0	0	1	32
neuro-hsc	13	0	1	0	14	448
cup-ecs	2	0	0	0	2	64
tid	0	0	1	0	1	32
biocomp	0	0	1	0	1	32
chakra	0	0	1	0	1	32
totals:	70	2	14	1	86	2752

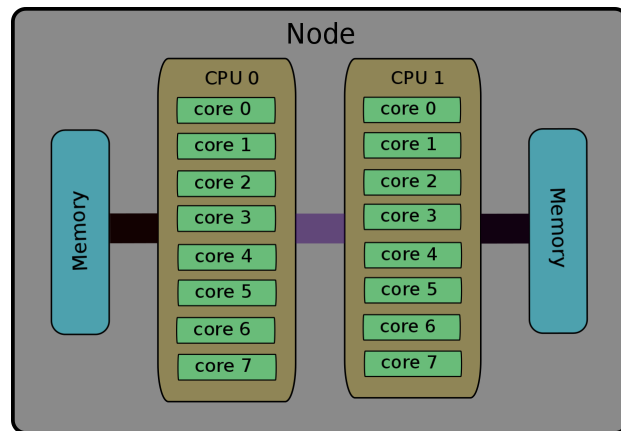
Compute nodes and partitions...



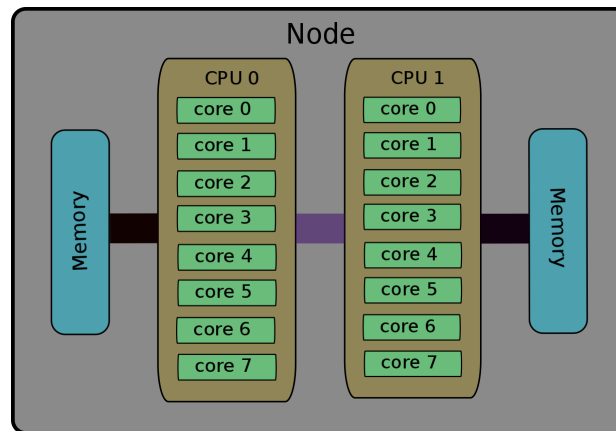
Head Node (wheeler)



Compute Nodes

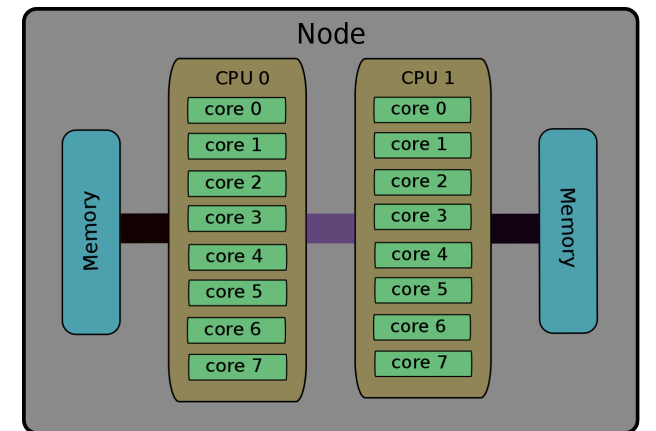


wheeler001



wheeler002

...



wheeler304

```
[vanilla@wheeler ~]$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                 8
On-line CPU(s) list:   0-7
Thread(s) per core:    1
Core(s) per socket:    4
Socket(s):              2
NUMA node(s):          2
Vendor ID:              GenuineIntel
CPU family:             6
Model:                  26
Model name:             Intel(R) Xeon(R) CPU X5550   @ 2.67GHz
```

Wheeler has 8 cores per node....

```
mfricke@hopper:~ $ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                64
On-line CPU(s) list:   0-63
Thread(s) per core:    2
Core(s) per socket:    16
Socket(s):             2
NUMA node(s):          2
Vendor ID:             GenuineIntel
CPU family:            6
Model:                85
Model name:            Intel(R) Xeon(R) Gold 6226R CPU @ 2.90GHz
```

Hopper has 32 real cores (64
virtual cores) per node....

Architecture

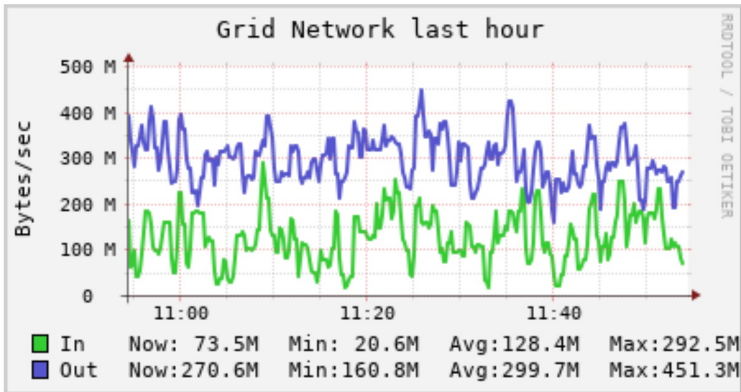
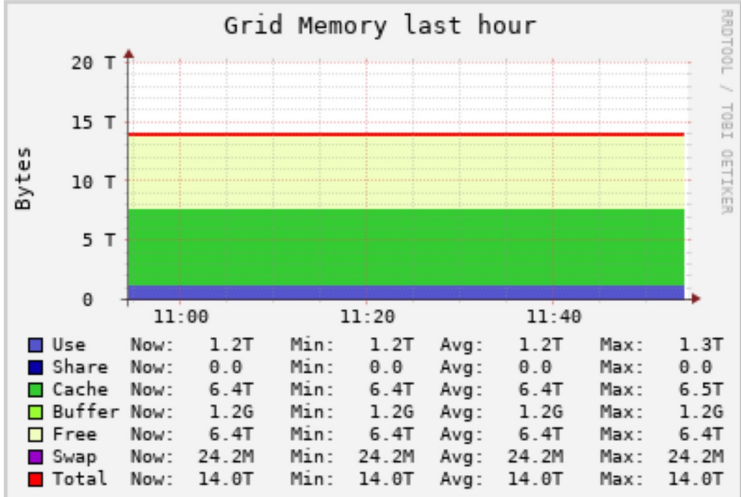
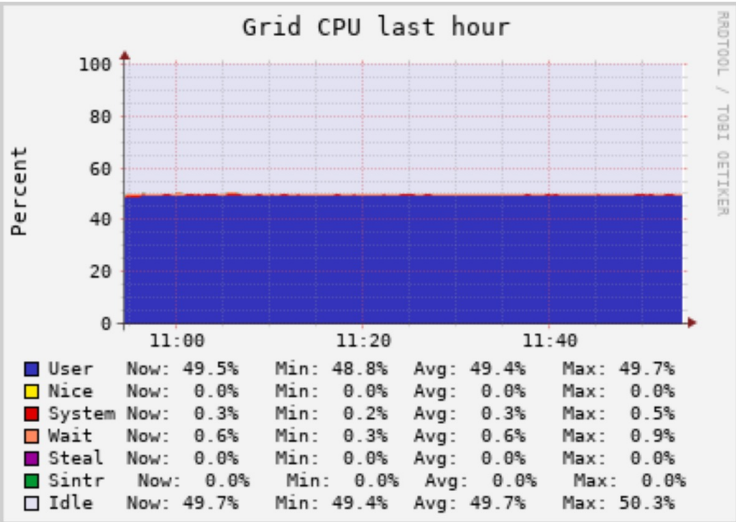
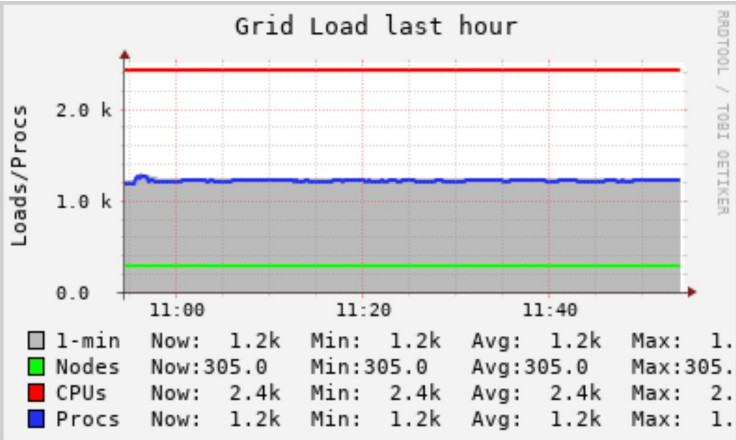
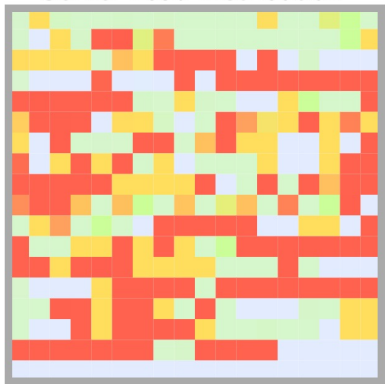
Wheeler Cluster Grid > Wheeler Cluster > --Choose a Node

Overview of @ 2022-03-03 18:54

CPU's Total: 2440
Hosts up: 305
Hosts down: 0

Current Load Avg (15, 5, 1m):
51%, 51%, 51%
Avg Utilization (last hour):
51%

Server Load Distribution





Technology, IT etc.

SLURM

means

Simple Linux Utility for Resource
Management

by acronymsandslang.com



VOTED **#1** SOFT DRINK OF THE 31ST CENTURY!



```
[vanilla@wheeler ~]$ squeue
```

	JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
	159914	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159915	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159916	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159917	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159918	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159919	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159912	normal	co-mcpdf	nsharma2	CG	2-00:00:28	1	wheeler257
	159913	normal	co-mcpdf	nsharma2	CG	2-00:00:28	1	wheeler257
166800_[21-100%10]		normal	Jannat	jannat	PD	0:00	1	(JobArrayTaskLimit)
	167067	normal	WINDENER	rubeldas	PD	0:00	36	
(QOSMaxCpuPerUserLimit)								
	167068	normal	WINDENER	rubeldas	PD	0:00	24	

Slurm....

```
[vanilla@wheeler ~]$ srun --partition debug --nodes 2 hostname  
srun: Account not specified in script or  
~/.default_slurm_account, using latest project  
wheeler302.alliance.unm.edu  
You have not been allocated GPUs. To request GPUs, use the -G  
option in your submission script.  
wheeler301.alliance.unm.edu  
[vanilla@wheeler ~]$ srun --partition debug --nodes 2 hostname
```

The srun command...

```
[vanilla@wheeler ~]$ srun --partition debug --ntasks 8 hostname  
srun: Account not specified in script or ~/.default_slurm_account, using  
latest project  
wheeler302.alliance.unm.edu  
wheeler302.alliance.unm.edu  
wheeler302.alliance.unm.edu  
wheeler302.alliance.unm.edu  
wheeler302.alliance.unm.edu  
wheeler302.alliance.unm.edu  
You have not been allocated GPUs. To request GPUs, use the -G option in your  
submission script.  
wheeler302.alliance.unm.edu  
wheeler302.alliance.unm.edu
```

The srun command...

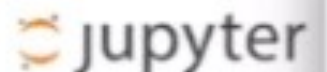
JupyterHub



python™

julia





Welcome to the

This Notebook Server was

WARNING

Don't rely on this serv

Your server is hosted thar

Run some Python

To run the code below:

1. Click on the cell to se
2. Press SHIFT+ENTER

A full tutorial for using the

In []: `!matplotlib inline`

```
import pandas as pd
import numpy as np
import matplotlib
```

Exploring the Lorenz System

In this Notebook we explore the [Lorenz system](#) of differential equations:

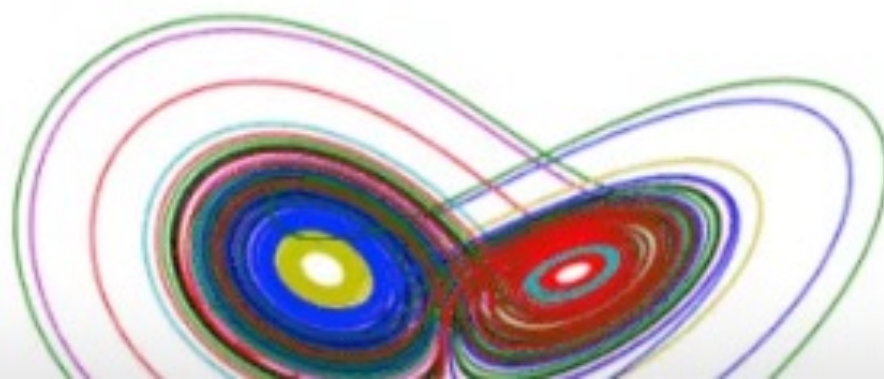
$$\dot{x} = \sigma(y - x)$$

$$\dot{y} = \rho x - y - xz$$

$$\dot{z} = -\beta z + xy$$

This is one of the classic systems in non-linear differential equations. It exhibits a range of complex behaviors as the parameters (σ, β, ρ) are varied, including what are known as chaotic solutions. The system was originally developed as a simplified mathematical model for atmospheric convection in 1963.

```
In [7]: interact(Lorenz, N=fixed(10), angle=(0.,360.),
                 sigma=(0.0,50.0), beta=(0.,5), rho=(0.0,50.0))
```



<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/index.html>



Description

Global Warming Science: A quantitative introduction to the science of climate change and its consequences, meant to assist students to process issues that often appear in the news and public debates. Topics include: the greenhouse effect and consequences of the rise of greenhouse gasses, including sea level rise, ocean acidification, heat waves, droughts, glacier melting, forest fires, expected changes to hurricanes and more. The scientific basis for each subject will be covered, and every class will involve a hands-on analysis of observations, climate models, and climate feedbacks, using python Jupyter notebooks. Throughout, an ability to critically evaluate observations, predictions and risk will be developed.

Sea level rise in the news



a submerged burial ground in Fiji's Togoru village. Five seawalls built to protect it have been knocked down by rising waters.

<https://www.nytimes.com/interactive/2014/03/27/world/climate-rising-seas.html>

St Mark's square in Venice on Nov 13, 2019. sea level rise of 1.87 m, highest in more than 50 yrs, flooding over 85% of city



Current sea level rise is ~3.5 mm/year...

<https://www.theguardian.com/environment/2019/dec/10/venice-floods-sea-level-rise-mose-project>

```
vanilla@hopper:~ $ cp -r /projects/shared/workshops/climate_change_jupyterhub ~  
vanilla@hopper:~ $
```

Dependencies

- The example code depends on these libraries:
- cartopy
- gsw
- Using it with jupyterhub requires ipykernel

```
kfotso@hopper:~$ module avail minicondas
```

```
----- /opt/spack/share/spack/modules/linux-rocky8-skylake_avx512 -----  
miniconda3-4.10.3-gcc-8.5.0-yikvdeb
```

```
----- /opt/spack/share/spack/modules/linux-rocky8-cascadelake -----  
miniconda3-4.10.3-gcc-11.2.0-tl4rbd6
```

```
----- /opt/spack/share/spack/lmod/linux-rocky8-x86_64/Core -----  
miniconda3/4.10.3-yikv
```

If the avail list is too long consider trying:

"module --default avail" or "ml -d av" to just list the default modules.

"module overview" or "ml ov" to display the number of modules for each name.

Use "module spider" to find all possible modules and extensions.

Use "module keyword key1 key2 ..." to search for all possible modules matching any of the "keys".

```
[kfotso@hopper:~$ module load miniconda3
```

```
[kfotso@hopper:~$ module list
```

Currently Loaded Modules:

1) miniconda3/4.10.3-yikv

Collecting package metadata (current_repodata.json): done

Solving environment: done

==> WARNING: A newer version of conda exists. <==
current version: 4.10.3
latest version: 4.14.0

Please update conda by running

```
$ conda update -n base -c defaults conda
```

Package Plan

environment location: /users/kfotso/.conda/envs/climate_change

[Proceed ([y]/n)? y

Preparing transaction: done

Verifying transaction: done

Executing transaction: done

"

```
kfotso@hopper:~$ source activate climate_change  
(climate_change) kfotso@hopper:~$
```

```
(climate_change) kfotso@hopper:~$ conda install cartopy ipykernel
Collecting package metadata (current_repodata.json): done
Solving environment: done
```

```
==> WARNING: A newer version of conda exists. <==
  current version: 4.10.3
  latest version: 4.14.0
```

Please update conda by running

```
$ conda update -n base -c defaults conda
```

```
## Package Plan ##
```

```
environment location: /users/kfotso/.conda/envs/climate_change
```

```
added / updated specs:
```

- cartopy
- ipykernel

The following NEW packages will be INSTALLED:

_libgcc_mutex	conda-forge/linux-64::_libgcc_mutex-0.1-conda_forge
_openmp_mutex	conda-forge/linux-64::_openmp_mutex-4.5-2_gnu

```
[(climate_change) kfotso@hopper:~$ conda install -c conda-forge gsw
Collecting package metadata (current_repodata.json): done
Solving environment: done
```

```
==> WARNING: A newer version of conda exists. <==
current version: 4.10.3
latest version: 4.14.0
```

Please update conda by running

```
$ conda update -n base -c defaults conda
```

```
## Package Plan ##
```

```
environment location: /users/kfotso/.conda/envs/climate_change
```

```
added / updated specs:
```

```
- gsw
```

The following NEW packages will be INSTALLED:

```
gsw                      conda-forge/linux-64::gsw-3.4.0-py310hde88566_3
```

Proceed ([y]/n)? ☐

Welcome to the Center for Advanced Research Computing

The UNM Center for Advanced Research Computing is the hub of computational research at UNM and one of the largest computing centers in the State of New Mexico. It is an interdisciplinary community that uses computational resources to create new research insights. The goal is to lead and grow the computational research community at UNM.

CARC provides not just the computing resources but also the expertise and support to help the university's researchers. This service is available to faculty, staff, and student researchers free of charge through support from the UNM Office of the Vice President for Research.

Get sta

Create a help ticket

Downtime notices

CARC systems information

CARC infrastructure

Resource limits

Storage and backup

Export control

JupyterHub cluster links

System status and downtime

System Usage by Principal Investigator

Hopper

Gibbs

Taos

Wheeler

Xena

JupyterHub cluster links:

Students go on to study greenhouse gas emissions

UNM Earth Science Professor to study the health of groundwater aquifers across the state

Explore or exploit: How our brains make choices



Sign in

Username:

Password:

Sign in

Server Options

Select a job profile:

- ✓

Debug Queue, 1 hours, 1 core, 4GB RAM
- Debug Queue, 4 hours, 8 cores, 24GB RAM
- General Access Queue, 48 hours, 32 cores, 90GB RAM
- Condo Queue (Preemptable), 48 hours, 32 cores, 90GB RAM
- Condo Queue (Preemptable), 48 hours, 32 cores, 90GB RAM, 1 GPU
- Chakra (restricted), 48 hours, 32 cores, 180GB RAM, 1 GPU
- Geodef (restricted), 1 hours, 1 core, 14GB RAM
- Geodef (restricted), 48 hours, 32 cores, 500GB RAM



You will be redirected automatically when it's ready for you.

Event log



☐  jupyterhub_hopper_slurm_cmd.err

☐  jupyterhub_hopper_slurm_cmd.out

☐  jupyterhub_hopper_batchspawner_28984.log

☐  jupyterhub_hopper_batchspawner_28982.log

☐  climate_change_jupyterhub

☐  Climate_course



[Files](#)[Running](#)[Clusters](#)

Select items to perform actions on them.



0

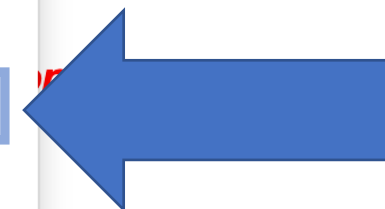
[/ Climate_course / Sea-level / code-for-students](#)[..](#)[sea_level_workshop.ipynb](#)[sea_level_variables.pickle](#)



Interrupt  
 Restart  
 Restart & Clear Output
 Restart & Run All
 Reconnect
 Shutdown

Change kernel ▶

Python 3
 Python [conda env:.conda-DarkMatter]
 Python [conda env:.conda-EQTransformer_final]
 Python [conda env:.conda-climate]
 Python [conda env:.conda-climate_change]
 Python [conda env:.conda-climatechange]
 Python [conda env:.conda-compatible_gpu]
 Python [conda env:.conda-eqt]
 Python [conda env:.conda-mpi_numpy]
 Python [conda env:.conda-taos_rgdal_import]
 R [conda env:.conda-taos_rgdal_import]



Global War

<https://courses.seas>

Courses/EPS101/

Sea Level

Please use the template below to answer

the notes

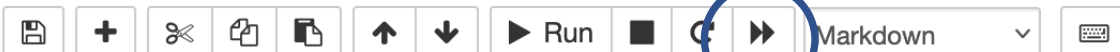
your name:

Required installations first:

from a terminal try:
 conda install cartopy
 pip install gsw

Another option for cartopy: in Anaconda Navigator, click Environments, then change in the pull-down menu "Installed" to "All"; find cart installation and click Apply.

If both of these options do not work, try "pip install cartopy" instead of "conda install cartopy" for cartopy.



If both of these options do not work, try `pip install cartopy` instead of `"conda install cartopy"` for cartopy.

```
In [1]: # import needed libraries and load data:
import numpy as np
import numpy.matlib as matlib
import pickle
import matplotlib.pyplot as plt
import scipy.integrate as integrate
import gsw
import cartopy.crs as ccrs
import cartopy.feature as cfeature
from cartopy.feature import NaturalEarthFeature
from cartopy import config
from matplotlib.colors import BoundaryNorm
from matplotlib import ticker
%matplotlib inline

# Read 2-d sea surface data for historical and rcp85 experiment
with open('./sea_level_variables.pickle', 'rb') as file:
    d = pickle.load(file)
    # print information about each extracted variable:
    for key in list(d.keys()):
        print("extracting pickled variable: name=", key, "; size=", d[key].shape)
        #print("type=", type(d[key]))
    globals().update(d)
```

```
extracting pickled variable: name= sealevel_lon ; size= (180, 360)
extracting pickled variable: name= sealevel_lat ; size= (180, 360)
extracting pickled variable: name= sealevel_historical ; size= (156, 180, 360)
extracting pickled variable: name= sealevel_historical_years ; size= (156,)
extracting pickled variable: name= sealevel_rcp85 ; size= (95, 180, 360)
```

Shut down your jupyterhub server

jupyterhub sea_level_workshop Last Checkpoint: an hour ago (autosaved)



Logout

Control Panel

File Edit View Insert Cell Kernel Widgets Help

Trusted

Python [conda env:.conda-climate] 



Global Warming Science

<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/>

Sea Level

Please use the template below to answer the *complete workshop questions from the course notes.*

your name:

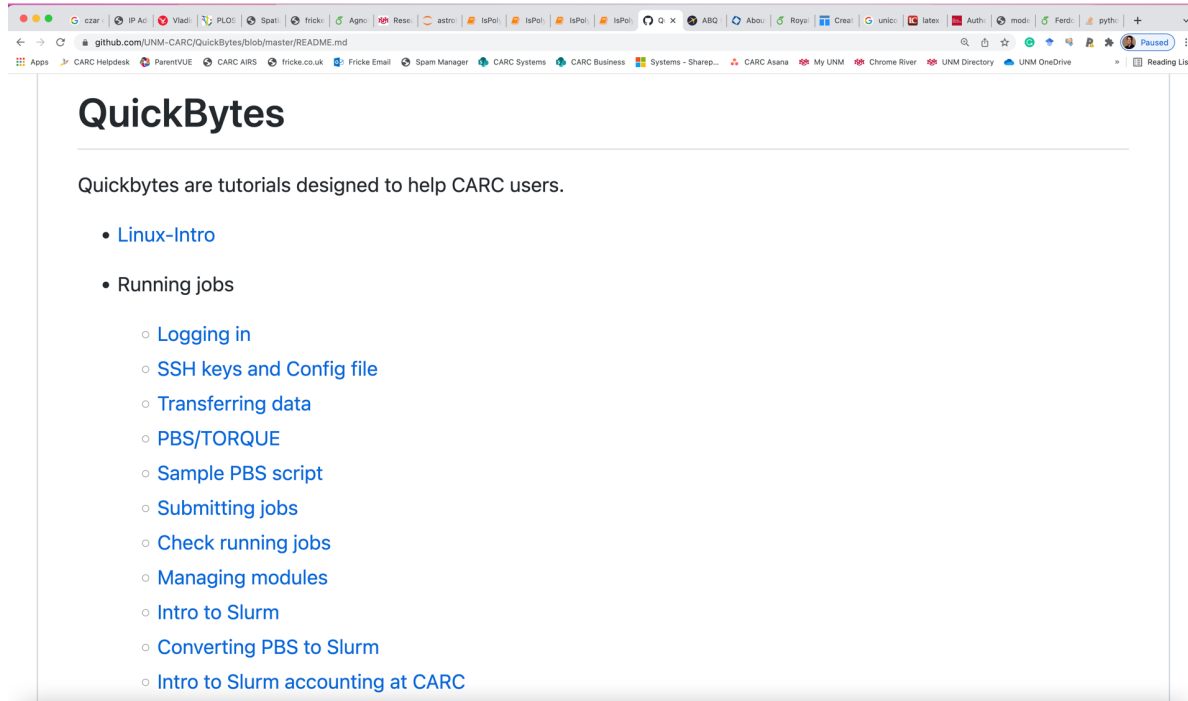
jupyterhub Home Token Admin

Stop My Server

My Server

CARC Resources

- Tutorial Videos
- Written Tutorials

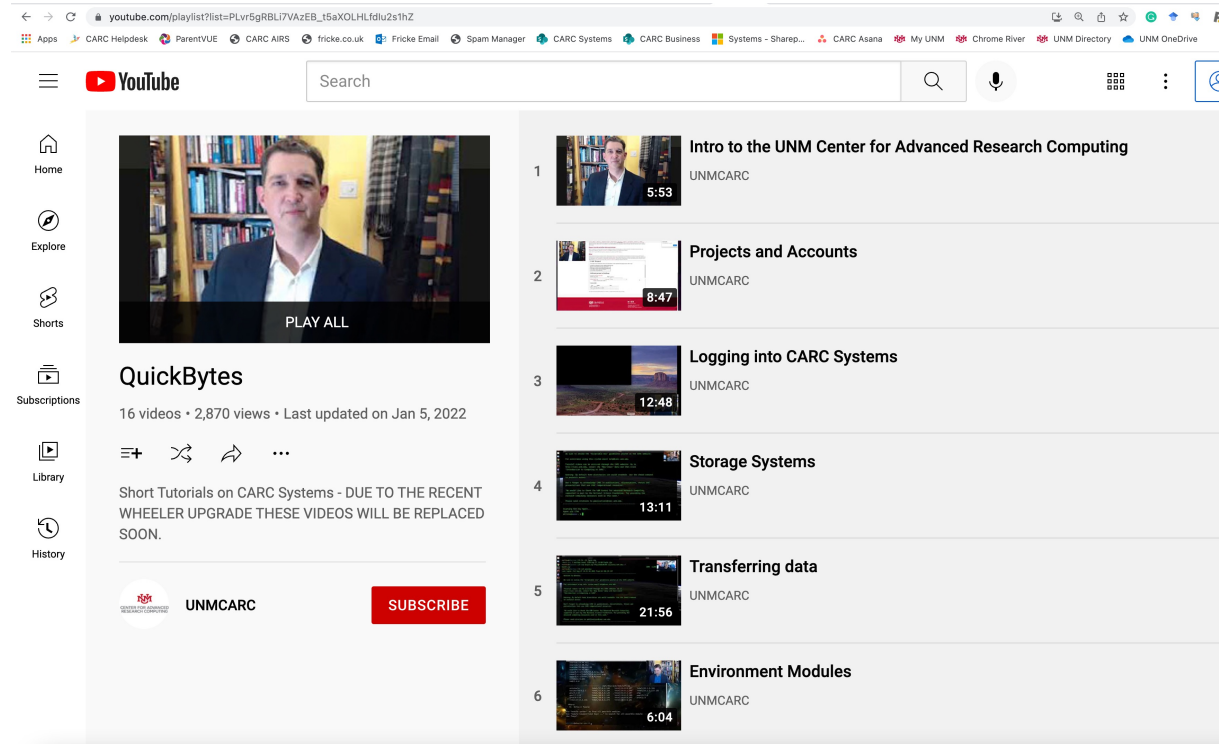


The screenshot shows a web browser window displaying the GitHub repository for UNMC-CARC/QuickBytes. The repository name is visible in the address bar. The page title is "QuickBytes". Below the title, a description states: "Quickbytes are tutorials designed to help CARC users." A list of tutorial topics is provided, including "Linux-Intro", "Running jobs", "Logging in", "SSH keys and Config file", "Transferring data", "PBS/TORQUE", "Sample PBS script", "Submitting jobs", "Check running jobs", "Managing modules", "Intro to Slurm", "Converting PBS to Slurm", and "Intro to Slurm accounting at CARC".

QuickBytes

Quickbytes are tutorials designed to help CARC users.

- [Linux-Intro](#)
- [Running jobs](#)
 - [Logging in](#)
 - [SSH keys and Config file](#)
 - [Transferring data](#)
 - [PBS/TORQUE](#)
 - [Sample PBS script](#)
 - [Submitting jobs](#)
 - [Check running jobs](#)
 - [Managing modules](#)
 - [Intro to Slurm](#)
 - [Converting PBS to Slurm](#)
 - [Intro to Slurm accounting at CARC](#)



The screenshot shows a YouTube playlist page for UNMC-CARC. The playlist is titled "QuickBytes" and contains 16 videos with 2,870 views, last updated on Jan 5, 2022. The video thumbnails and titles are as follows:

1. [Intro to the UNM Center for Advanced Research Computing](#) (5:53)
2. [Projects and Accounts](#) (8:47)
3. [Logging into CARC Systems](#) (12:48)
4. [Storage Systems](#) (13:11)
5. [Transferring data](#) (21:56)
6. [Environment Modules](#) (6:04)

The UNMC-CARC logo and a "SUBSCRIBE" button are visible at the bottom of the video player area.

Getting Help

help@carc.unm.edu

Office hours